SHAKTI

A small dive into Dhrystone & CoreMark

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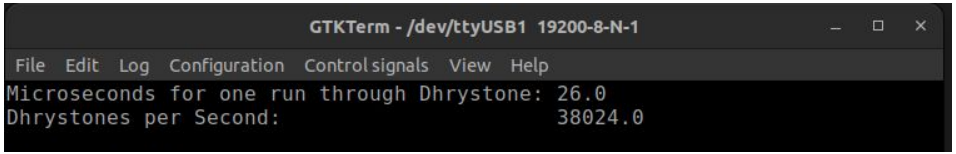
What is Dhrystone?

- Dhrystone is a free, synthetic benchmark used to measure general-purpose processor performance, particularly in embedded systems.
- Created in 1984 by Reinhold Weicker, it became an industry-standard benchmark for evaluating integer and string operation performance before being succeeded by CoreMark.

Let us Dhrystone

Command to run

```
$ make software PROGRAM=dhrystone TARGET=yamuna
```



How does Dhrystone work?

- **Workload Includes:**
  - String handling and manipulation (character and memory operations).
  - Integer arithmetic and logic operations.
  - Control flow (loops, conditionals, and procedure calls).
  - Struct and pointer operations simulating system-level workloads.
  - Portable: Written in C, ensuring compatibility across platforms.
- **Key Metrics:**
  - **Dhrystones/Second**: Number of times the benchmark loop executes per second.
  - **DMIPS (Dhrystone MIPS):** Performance metric normalized against a VAX 11/780 computer (1 DMIPS = 1757 Dhrystones/second).

Scores of popular CPUs

CPU	Clock (MHz)	Dhrystone 2 Opt (VAX)	DMIPS/MHz
AMD 80386	40	13.7	0.34[1]
Pentium 100	100	122	1.22[1]
Pentium III 1000	1000	1595	1.60[1]
Pentium 4 3066	3066	4012	1.31[1]
Core i5 2467M	2467	4752	1.93[1]

Sources: *[1]R. Longbottom, "Dhrystone Benchmark Results on PCs and Later Devices," Online – ResearchGate, 2017. Available: https://www.researchgate.net/publication/319176416_Dhrystone_Benchmark_Results_On_PCs_and_Later_Devices_Roy_Longbottom*

Dhrystone

Weicker, R. P. (1984). Dhrystone: A synthetic systems programming benchmark. Communications of the ACM, 27(10), 1013–1030. https://doi.org/10.1145/358274.358283

Should we use Dhrystone?

Pros

- **Standardization**: Provides a consistent metric for comparing integer performance across processors.
- **Simplicity**: Small, portable C code (~2,000 lines for Dhrystone 2) that is easy to compile and run.
- **Transparency**: Widely published results and open access make it easy to verify and reproduce.
- **Historical Benchmark**: Long history allows comparison across decades of CPUs.

Cons

- **Narrow Focus**: Measures mostly integer arithmetic and string/logic operations; ignores floating-point, memory/cache performance, I/O, and multicore scaling.
- **Synthetic Workload**: Does not necessarily reflect real-world application performance; results can be heavily affected by compiler optimizations.
- **Obsolete for Modern CPUs**: Modern superscalar, out-of-order, and multi-core architectures are not fully stressed by Dhrystone.
- **Compiler Sensitivity**: Scores can vary significantly depending on compiler version and optimization flags.

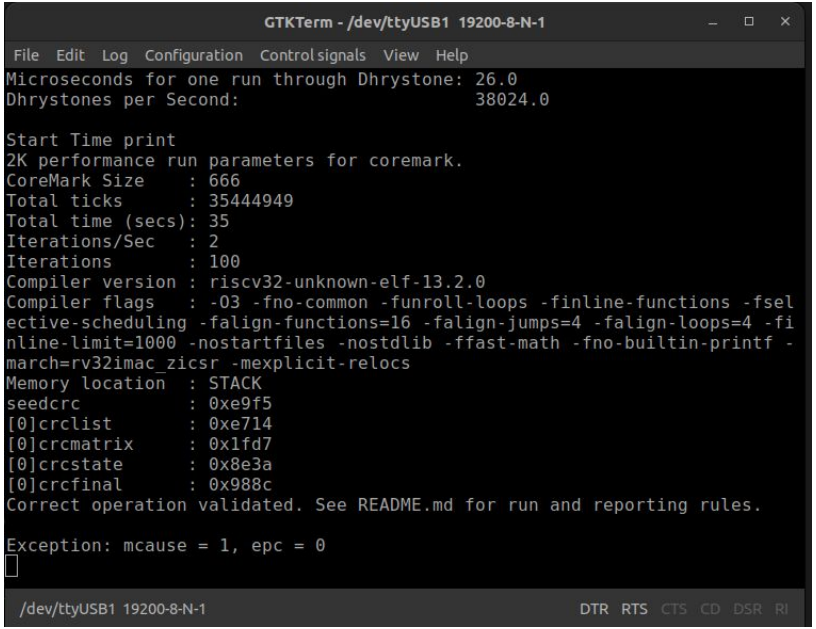
What is CoreMark?

- CoreMark is a free, industry-standard benchmark for measuring embedded processor performance.
- Created by the Embedded Microprocessor Benchmark Consortium (EEMBC) to replace Dhrystone.

Let us CoreMark

Command to run

```
$ make software PROGRAM=coremarks TARGET=yamuna
```



How does CoreMark work?

- **Workload Includes**:
  - Linked list traversal (memory access).
  - Matrix multiplication (integer operations).
  - State machine (branching logic).
  - CRC (cyclic redundancy check for data integrity).
- **Portable**: Written in C, ensuring compatibility across platforms.
- **Key Metrics**:
  - **CoreMarks=Iterations/Second**: Number of times the benchmark runs per second.
  - **CoreMark/MHz**: Normalized score accounting for clock speed, enabling cross-architecture comparisons.

Scores of popular CPUs

CPU	Core Count	Clock Frequency (GHz)	Score (CM/MHz)
i7 10510U	4	4.9	22.96[1]
i5 12500H	16	4.5	71.95[1]
Ryzen 9 7900X	12	4.7	176.8[1]
SiFive HiFive U	4	1.5	2.01[1]
ARM Cortex A53	4	1.5	13.12[1]
SonicBOOM	1	3.2	6.2[2]
Rocket	1	1.5	2.32[2]

***Sources**: [1] EEMBC, [2] Krste Asanovic, David A Patterson, and Christopher Celio. 2015. The Berkeley out of-order machine (boom): An industry-competitive, synthesizable, parameterized RISC-V processor. Technical Report. University of California at Berkeley, Berkeley, United States.*

Should we use CoreMark?

1. **Pros**:
   a. **Standardization**: Enables apples-to-apples comparisons between processors.
   b. **Simplicity**: Small code footprint (~1,200 lines) and easy to run.
   c. **Transparency**: Open-source and free to use.
2. **Cons**:
   a. **Narrow Focus**: Only focuses on pipeline efficiency and does not measure memory/cache performance, I/O, power consumption, or multicore scaling.
   b. **Synthetic**: May not reflect real-world application performance.

Thank you

Website: shakti.org.in

GitLab: gitlab.com/shaktiproject/cores/c-class